

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

|                  |                                                                                                              |               |                                             |
|------------------|--------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------|
| Course Code:     | CSA0305                                                                                                      | Course Title: | Data Structures                             |
| CO Assessed:     | CO5 – Naturalization (BL4, BL6)                                                                              | Assessment:   | Assessment Tool 1 – Pseudocode Writing Task |
| Weightage:       | 35% of CIA                                                                                                   | Total Marks:  | 100                                         |
| CO5 Statement:   | Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem. |               |                                             |
| Student Name:    | A. Gopika                                                                                                    | Reg. No.:     | 192521343                                   |
| Section / Batch: | B. Tech IT                                                                                                   | Date:         | 31/08/2026                                  |

### Code of Conduct

*I A. Gopika/192521343 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: A. Gopika

### Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic         | Focus                         | Q Nos | Marks     |
|-------------------------|-------------------------------|-------|-----------|
| Minimum Spanning Tree   | Kruskal's Algorithm           | 1     | 20        |
| Graph Sorting           | Topological Sort              | 2     | 20        |
| Shortest Path Algorithm | MST & Dijkstra's Algorithm    | 3     | 20        |
| Shortest Path Algorithm | Dijkstra's Algorithm          | 4     | 20        |
| Minimum Spanning Tree   | Prim's or Kruskal's Algorithm | 5     | 20        |
| GRAND TOTAL:            |                               |       | 100 Marks |

## Rubrics for Calculation

| Criteria                              | Weightage | Level 4 – Exemplary                                                           | Level 3 – Proficient                                 | Level 2 – Developing                            | Level 1 – Beginning               |
|---------------------------------------|-----------|-------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------|-----------------------------------|
| <b>Problem Understanding</b>          | 20        | Demonstrates complete and clear understanding; handles edge cases effectively | Good understanding; minor gaps in edge case handling | Partial understanding; misses some requirements | Misinterprets problem; major gaps |
| <b>Code Correctness</b>               | 30        | Fully correct; passes all test cases                                          | Mostly correct; minor errors                         | Partially correct; several test cases fail      | Incorrect or non-functional code  |
| <b>Code Quality &amp; Readability</b> | 20        | Clean, modular, well-commented, follows standards                             | Readable with some structure                         | Limited readability; poor structure             | Unorganized and hard to read      |
| <b>Complexity Analysis</b>            | 20        | Accurate time & space complexity with justification                           | Correct but limited explanation                      | Incomplete or partially correct                 | Missing or incorrect              |
| <b>Testing &amp; Validation</b>       | 10        | Thorough testing including edge cases                                         | Adequate testing with few edge cases                 | Minimal testing                                 | No testing evidence               |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## QUESTIONS

**Q1** Minimum Spanning Tree - Kruskal's Algorithm  
20 marks

**Q2** Graph - Topological Sort  
20 marks

**Q3** Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's  
20 marks

**Q4** Shortest Path Algorithm - Dijkstra's Algorithm  
20 marks

**Q5** Minimum Spanning Tree - Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

### Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

#### Your Answer

```
BEGIN
Read v, E
For each vertex i=1 to v
  Makeset(i)
sort all edges in increasing order of weight
MST = empty
```

Save Answer

Next →

## QUESTIONS

**Q1** Minimum Spanning Tree - Kruskal's Algorithm  
20 marks

**Q2** Graph - Topological Sort  
20 marks

**Q3** Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's  
20 marks

**Q4** Shortest Path Algorithm - Dijkstra's Algorithm  
20 marks

**Q5** Minimum Spanning Tree - Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

### Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- Detect cycle before sorting
- Use DFS approach

#### Your Answer

```
Topologicalsort(G);
visited=[]
recstack=[]
stack=empty
```

for each vertex v in G:

Save Answer

Next →

## QUESTIONS

**Q1** ✓  
Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

**Q2** ✓  
Graph - Topological Sort  
20 marks

**Q3** ✓  
Shortest Path Algorithm -  
Minimum Spanning Tree &  
Dijkstra's  
20 marks

**Q4** ✓  
Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

**Q5** ✓  
Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

### Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- Use MST for infrastructure
- Use shortest path for routing

#### Your Answer

Algorithm Smart Transportation(Graph G, Source S, Destination D)

Input:

G=Road network(vertices, Edges with weights)  
s=source city  
D=Destination city

Save Answer

Next →

## QUESTIONS

**Q1** ✓  
Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

**Q2** ✓  
Graph - Topological Sort  
20 marks

**Q3** ✓  
Shortest Path Algorithm -  
Minimum Spanning Tree &  
Dijkstra's  
20 marks

**Q4** ✓  
Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

**Q5** ✓  
Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

### Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

#### Your Answer

Algorithm Dijkstra with voucher(Graph G,Source s, Destination D)

Input:

G=Weighted graph  
S=Source  
D=Destination

Save Answer

Next →

## QUESTIONS

**Q1** ✓  
Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

**Q2** ✓  
Graph - Topological Sort  
20 marks

**Q3** ✓  
Shortest Path Algorithm -  
Minimum Spanning Tree &  
Dijkstra's  
20 marks

**Q4** ✓  
Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

**Q5** ✓  
Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

5/5 answered

### Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.

Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

#### Your Answer

Algorithm SecondBest MST(Graph G)

1. MST=Kruskal(G)
2. Bestcost=cost(MST)
3. SecondBest =INF
4. For each edge e in MST

Save Answer